Parsing in NLP

Introduction to Parsing

Parsing is an important technique in **Natural Language Processing (NLP)** used to analyze the **grammatical structure of a sentence**.

When humans read a sentence, we easily understand **who is doing the action and what the action is**.
However, computers cannot understand natural language directly. Parsing helps the computer **break a sentence into smaller parts and understand relationships between words**.

In simple words:

Parsing = Analyzing a sentence to identify its grammatical structure.

Example sentence:
"The boy plays football."

Parsing identifies:

- Subject → *boy*
- Verb → *plays*
- Object → *football*

So, the system understands the meaning of the sentence.

Concept of Parsing

The main concept of parsing is to **convert a sentence into a structured representation**.

A sentence usually contains:

- **Noun Phrase (NP)** – person or thing
- **Verb Phrase (VP)** – action
- **Prepositional Phrase (PP)** – location or relation

Example sentence:

“The girl reads a book.”

Structure: Sentence (S)

- Noun Phrase (NP) → The girl
- Verb Phrase (VP) → reads a book

This structure is usually represented using a **Parse Tree**.

How Parsing Works in NLP

Parsing normally follows these steps.

Step 1 – Input Sentence

Example:
“Students study NLP.”

Step 2 – Tokenization

Sentence is split into words.

Students | study | NLP

Step 3 – Part-of-Speech Tagging

Each word gets a grammatical tag.

Students → Noun
study → Verb
NLP → Noun

Step 4 – Build Parse Structure

The parser organizes the words according to grammar rules.

Sentence

- Subject → Students
- Verb → study
- Object → NLP

Example:

Question: Given Sentence - “the big dog chased the cat”

Grammar Rules

S → NP VP
NP → Det N | Det Adj N
VP → V NP | V
Det → the
Adj → big
N → dog | cat
V → chased

Answer:

1. Sentence Breakdown

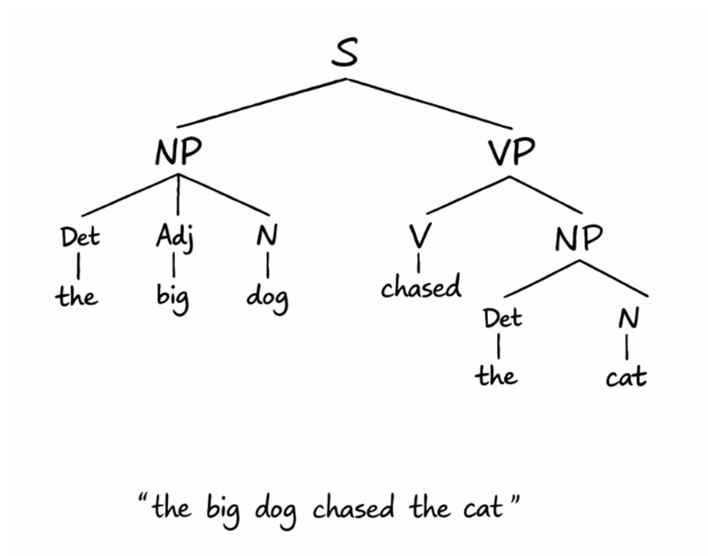
The | big | dog | chased | the | cat

2. Derivation

1. S ⇒ NP VP
2. S ⇒ Det Adj N VP
3. S ⇒ the big dog VP
4. S ⇒ the big dog V NP
5. S ⇒ the big dog chased NP
6. S ⇒ the big dog chased Det N
7. S ⇒ the big dog chased the cat

3. Final Derived Sentence

the big dog chased the cat



Applications of Parsing

1. **Machine Translation** – helps translate sentences correctly by understanding grammar.
2. **Information Extraction** – identifies important data like subject, object, names, etc.
3. **Question Answering Systems –** understands questions to give accurate answers.
4. **Speech Recognition** – helps convert and interpret spoken language correctly.